HACKATHON PROJECT · 2026

Xscholar

An autonomous AI research intelligence agent that scrapes live papers, builds a cross-reference knowledge graph, and deploys itself — zero human intervention.

OPENCLAW AGENT · TIDB CLOUD · MEM9 MEMORY · LIVE DATA · AUTO-DEPLOY

http://178.104.103.88:3000 · github.com/yx-yan/xscholar

THE PROBLEM

Research is drowning in papers.

100+ new papers drop daily on arXiv alone. Researchers have no tool that autonomously tracks, filters, cross-references, and *remembers* what matters — across sessions, across time.

Manual overload

Researchers manually scan feeds, bookmark papers, lose context between sessions.

No cross-reference

Papers are read in isolation. No tool connects today's paper to last month's breakthrough.

Expensive tools

Existing solutions are paywalled, closed-source, or require manual curation.

THE SOLUTION

An agent that thinks like a researcher.

Xscholar is an OpenClaw skill — an autonomous agent that runs 24/7, fetches live papers, builds a semantic knowledge graph, and answers questions in natural language.

INTELLIGENCE

Doesn't just store papers — cross-references them, detects emerging research threads, builds a living momentum summary.

AUTONOMY

Fully scheduled via OpenClaw cron. No manual triggers. Wakes up, fetches, indexes, notifies — then goes back to sleep.

MEMORY

Two persistence layers: TiDB Cloud for structured SQL queries, mem9 for semantic vector search across all sessions.

OPEN SOURCE

MIT licensed. Installable as an OpenClaw skill by anyone. Configure your research profile and it's running in minutes.

ARCHITECTURE

Three-layer intelligence stack

LAYER 1 · SCRAPING

Paper Pipeline

fetch-papers.js pulls live papers from arXiv & Semantic Scholar. Scores each paper 0.0–1.0 against your research profile keywords.

ARXIV API · SEMANTIC SCHOLAR

→

LAYER 2 · STORAGE

Dual Persistence

TiDB Cloud Zero stores structured paper data (SQL). mem9 stores intelligence — connections, threads, summaries — as semantic vectors.

TIDB CLOUD · MEM9 API

→

LAYER 3 · INTERFACE

Agent + UI

OpenClaw agent answers research questions in natural language. Next.js star-field UI visualizes papers on a live, spinning night sky.

OPENCLAW · NEXT.JS

LIVE & RUNNING

Real data. Right now.

20+
PAPERS INDEXED
FROM LIVE ARXIV

2
PERSISTENCE LAYERS
TIDB + MEM9

24/7
AUTONOMOUS AGENT
VIA OPENCLAW CRON

```
# Agent answers research questions in natural language node scripts/query.js --q "diffusion models for segmentation" --limit 5 # Output:
live JSON from TiDB Cloud [{ "title": "RefAlign: Representation Alignment...", "relevance": 0.4, "source": "arxiv" }, { "title":
"PixelSmile: Toward Fine-Grained...", "relevance": 0.2, "source": "arxiv" }]
```

VISUAL INTERFACE

Papers as a night sky.

Every paper is a star. Brightness = relevance. Color = research topic. The sky spins, zooms, and responds to touch.

Star Field Canvas

WebGL-style canvas. Auto-rotating, zoomable. Papers orbit by topic cluster.

Smart Sidebar

Keyword search, relevance slider, date filter. Results update the star field live.

DB Dashboard

Live stats: total papers, 24h ingestion, top topics, source breakdown — all from TiDB.

Live demo: http://178.104.103.88:3000

JUDGING CRITERIA

Built for functionality, not theatre.

Scraping ✓

fetch-papers.js pulls live data from arXiv and Semantic Scholar APIs on every run. No mocked data — actual papers from today.

Storage ✓

setup.js autonomously creates the DB schema in TiDB Cloud Zero. Papers persist across restarts. mem9 stores intelligence layer.

Processing ✓

index-to-mem9.js cross-references new papers against existing knowledge, detects research threads, updates momentum summary.

Deployment ✓

Next.js UI deployed on the same server via systemd + Kubernetes ingress. Public URL. Running right now, not just localhost.

Xscholar

The research intelligence layer your OpenClaw agent was missing.

LIVE DEMO
http://178.104.103.88:3000

SOURCE CODE
github.com/yx-yan/xscholar

OPENCLAW AGENT · TIDB CLOUD ZERO · MEM9 MEMORY · ARXIV · SEMANTIC SCHOLAR · NEXT.JS · NODE.JS

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